Pearson Edexcel Level 3 GCE

Friday 16 May 2025

Afternoon

Paper reference **8FM0/28**

Further Mathematics

Advanced Subsidiary
Further Mathematics options
28: Decision Mathematics 2
(Part of option K only)

You must have:
Mathematical Formulae and Statistical Tables (Green), calculator,
D2 Answer Book (enclosed)

Candidates may use any calculator allowed by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

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Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
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Turn over ▶

P 7 4 0 7 6 A

Pearson

1. Five workers, A, B, C, D and E, are each to be assigned to one of five tasks, J, K, L, M and N. Each task must be assigned to exactly one worker and each worker must do exactly one task.

Worker C cannot do task L and worker D cannot do task K.

The profit, in pounds, that each worker will make while assigned to each task is shown in the table below.

	J	K	L	M	N
A	38	33	40	35	32
B	26	24	27	25	23
C	33	29	–	30	27
D	36	–	41	37	33
E	32	27	31	29	25

The Hungarian algorithm is to be used to find the maximum total profit that can be earned by the five workers.

(a) Explain how the contents of the table must be modified to allow the algorithm to be used.

(2)

(b) Reducing rows first, use the Hungarian algorithm to obtain the maximum total profit. You should explain how any initial row and column reductions are made and also how you determine if the table is optimal at each stage.

(7)

(Total for Question 1 is 9 marks)

2.

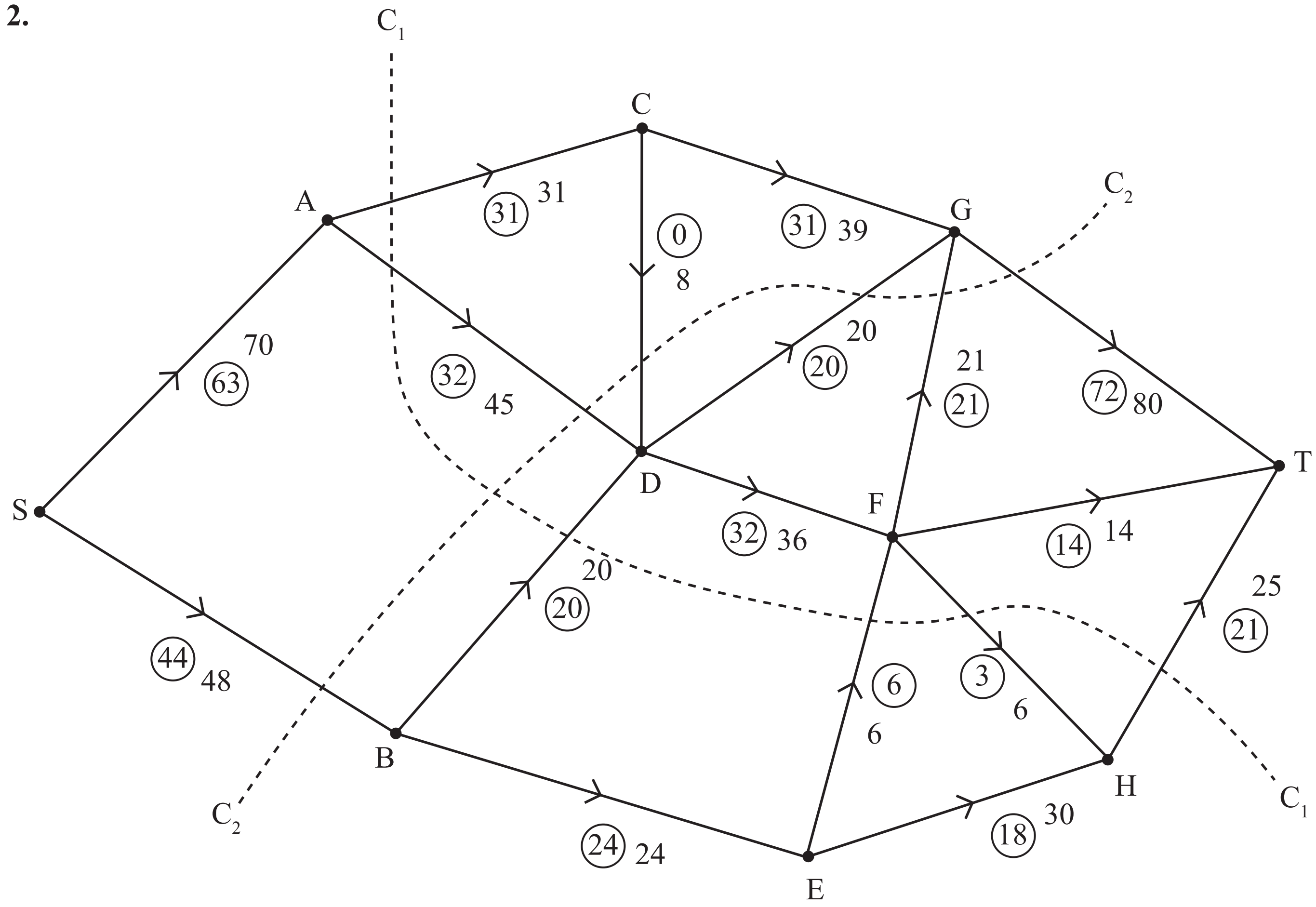

Figure 1

Figure 1 shows a capacitated, directed network of pipes. The number on each arc represents the capacity of the corresponding pipe. The numbers in circles represent a feasible flow from S to T.

(a) State the two conditions satisfied by a feasible flow.

(2)

(b) List the seven saturated arcs in Figure 1.

(1)

(c) Find the capacity of

(i) cut C_1

(ii) cut C_2

(2)

(d) Write down a flow-augmenting route that increases the flow by four units.

(1)

(e) Use the answer to part (d) to draw the resulting flow pattern on Diagram 1 in the answer book.

(2)

(f) Prove that the answer to part (e) is a maximum flow.

(3)

(Total for Question 2 is 11 marks)

3. Layla and Mohsin play a zero-sum game represented by the following pay-off matrix for Layla.

	Mohsin plays X	Mohsin plays Y	Mohsin plays Z
Layla plays P	1	−2	2
Layla plays Q	−4	3	−5
Layla plays R	−1	1	−3

(a) (i) Find the play-safe strategies for each player.

(ii) State, giving a reason, whether there is a stable solution to this game.

(3)

(b) Option R is now removed from Layla's choices.

(i) For each of Mohsin's three options, find the expected pay-off to Layla when she plays options P and Q equally often.

(1)

(ii) Use a graphical method to determine Layla's optimal mixed strategy. You should define any variables you use.

(7)

(Total for Question 3 is 11 marks)

---

4. A sequence $\{u_n\}$, where $n \geqslant 1$, satisfies the recurrence relation

$$4u_{n+1} + 2u_n = 3n - 5$$

Given that $u_1 = -2$

(a) solve the recurrence relation, giving u_n in terms of n

(6)

(b) hence determine the number of negative terms in the sequence $\{u_n\}$. You must justify your answer.

(3)

(Total for Question 4 is 9 marks)

TOTAL FOR DECISION MATHEMATICS 2 IS 40 MARKS

Pearson Edexcel Level 3 GCE

Friday 17 May 2024

Afternoon

Paper reference **8FM0/28**

Further Mathematics

Advanced Subsidiary
Further Mathematics options
28: Decision Mathematics 2
(Part of option K only)

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Turn over

Pearson

1.

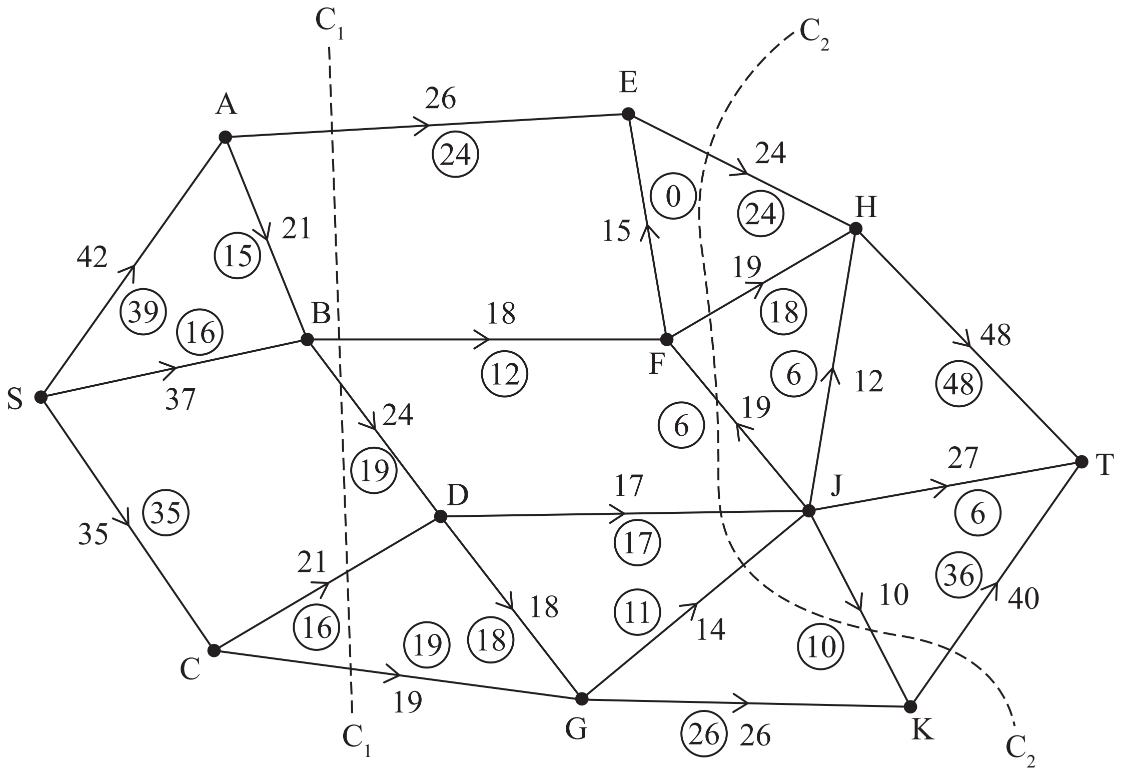

Figure 1

Figure 1 shows a capacitated, directed network of pipes. The number on each arc represents the capacity of the corresponding pipe. The numbers in circles represent a feasible flow from S to T.

(a) State the value of this flow.

(1)

(b) Explain why arcs CD and CG cannot both be saturated.

(1)

(c) Find the capacity of

(i) cut C_1

(ii) cut C_2

(2)

(d) Write down a flow augmenting route of weight 6 which saturates BF.

(1)

The flow augmenting route in part (d) is applied to give an increased flow.

(e) Prove that this increased flow is maximal.

(3)

(Total for Question 1 is 8 marks)

2. A team of 5 players, A, B, C, D and E, competes in a quiz. Each player must answer one of 5 rounds, P, Q, R, S and T.

Each player must be assigned to exactly one round, and each round must be answered by exactly one player.

Player B cannot answer round Q, player D cannot answer round T, and player E cannot answer round R.

The number of points that each player is expected to earn in each round is shown in the table.

	P	**Q**	**R**	**S**	**T**
A	32	40	35	41	37
B	38	–	40	27	33
C	41	28	37	36	35
D	35	33	38	36	–
E	40	38	–	39	34

The team wants to maximise its total expected score.

The Hungarian algorithm is to be used to find the maximum total expected score that can be earned by the 5 players.

(a) Explain how the table should be modified.

(2)

(b) (i) Reducing rows first, use the Hungarian algorithm to obtain an allocation which maximises the total expected score.

(ii) Calculate the maximum total expected score.

(6)

(Total for Question 2 is 8 marks)

3. Haruki and Meera play a zero-sum game. The game is represented by the following pay-off matrix for Haruki.

		Meera		
		Option X	Option Y	Option Z
Haruki	Option A	4	–2	–5
	Option B	1	4	–3
	Option C	–1	6	1
	Option D	–4	5	3

(a) Determine whether the game has a stable solution.

(2)

Option Y for Meera is now removed.

(b) Write down the reduced pay-off matrix for Meera.

(1)

(c) (i) Given that Meera plays Option X with probability p, determine her best strategy.

(ii) State the value of the game to Haruki.

(iii) State which option(s) Haruki should never play.

(7)

The number of points scored by Haruki when he plays Option C and Meera plays Option X changes from –1 to k

Given that the value of the game is now the same for both players,

(d) determine the value of k. You must make your method and working clear.

(4)

(Total for Question 3 is 14 marks)

4. Peter sets up a savings plan. He makes an initial deposit of £D and then pays in £M at the end of each month.

The value of the savings plan, in pounds, is modelled by

$$u_{n+1} = 1.025\, u_n + 1800$$

where $n \geqslant 0$ is an integer and u_n is the total value of the savings plan, in pounds, after n years.

(a) Calculate the value of M

(1)

Given that the value of the savings plan after 1 year is £6925

(b) solve the recurrence relation for u_n

(5)

(c) Determine the value of D

(1)

(d) Hence determine, using algebra, the number of years it will take for the value of the savings plan to exceed £20000

(3)

(Total for Question 4 is 10 marks)

TOTAL FOR DECISION MATHEMATICS 2 IS 40 MARKS

Pearson Edexcel Level 3 GCE

Friday 19 May 2023

Afternoon

Paper reference **8FM0/28**

Further Mathematics

Advanced Subsidiary
Further Mathematics options
28: Decision Mathematics 2
(Part of option K only)

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D2 Answer Book (enclosed)

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Advice

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- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ▶

1. Five workers, A, B, C, D and E, are available to complete four tasks, P, Q, R and S.

Each worker can only be assigned to at most one task, and each task must be done by at most one worker.

Worker B cannot be assigned to task Q and worker E cannot be assigned to task S.

The time, in minutes, that each worker takes to complete each task is shown in the table below.

	P	Q	R	S
A	38	39	37	37
B	39	–	39	40
C	41	44	40	42
D	40	41	39	38
E	36	39	41	–

The Hungarian algorithm is to be used to find the least total time to complete all four tasks.

(a) Explain how the table should be modified so that the Hungarian algorithm can be applied.

(2)

(b) (i) Use the Hungarian algorithm to obtain an allocation that minimises the total time.

(ii) Explain how you determined if the table was optimal at each stage.

(6)

(c) Calculate the least total time to complete all four tasks.

(1)

(Total for Question 1 is 9 marks)

2.

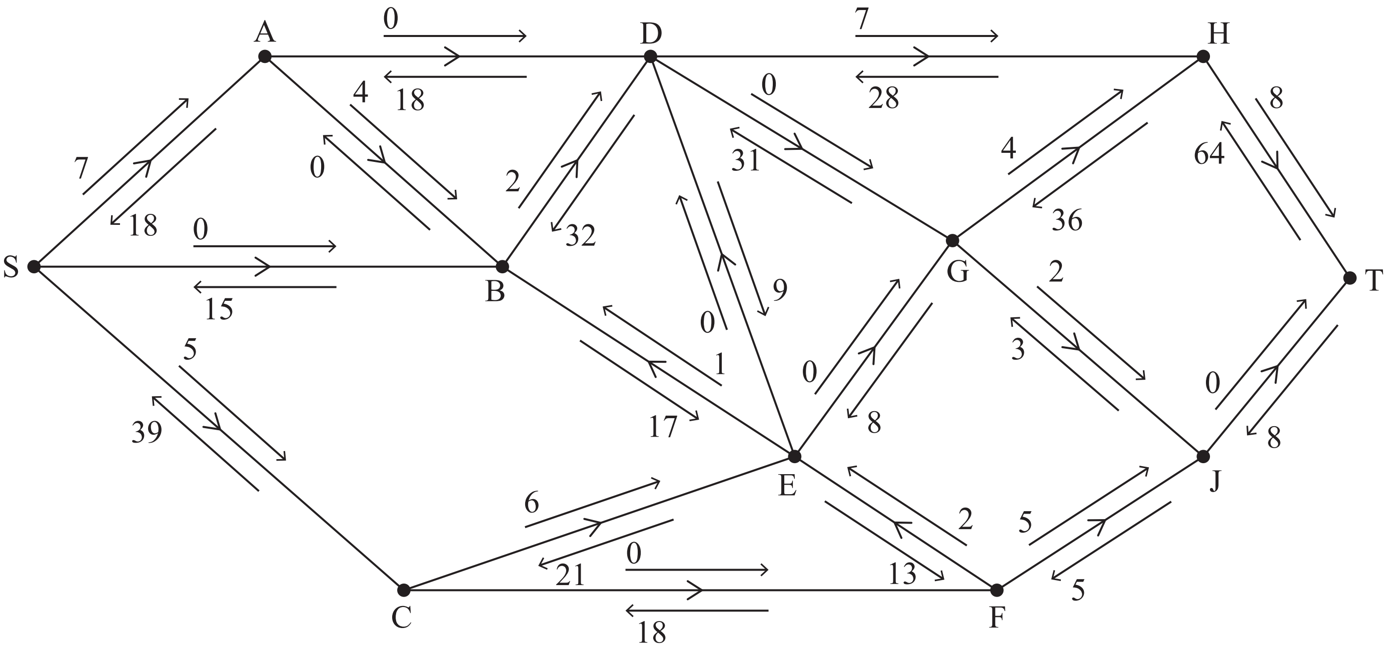

Figure 1

An engineer monitors a system of pipes through which a fluid flows from the source, S, to the sink, T.

The engineer initialises the labelling procedure for this system, and the excess capacities and potential backflows are shown on the arrows either side of each arc, as shown in Figure 1.

(a) State the value of the initial flow.

(1)

(b) Obtain the capacity of the cut that passes through the arcs SA, SB, CE, FE and FJ.

(1)

(c) Use the labelling procedure to find a maximum flow through the network. You must list each flow-augmenting route you use, together with its flow.

(3)

(d) Use your answer to (c) to draw a maximum flow pattern on Diagram 1 in the answer book.

(1)

(e) Prove that the answer to (d) is optimal.

(3)

(Total for Question 2 is 9 marks)

3. A two-person zero-sum game is represented by the following pay-off matrix for player A.

	B plays X	B plays Y
A plays Q	2	–2
A plays R	–1	5
A plays S	3	4
A plays T	0	2

(a) (i) Show that this game is stable.

(ii) State the value of this game to player B.

(3)

Option S is removed from player A's choices and the reduced game, with option S removed, is no longer stable.

(b) Write down the reduced pay-off matrix for player B.

(1)

Let B play option X with probability p and option Y with probability $1 - p$.

(c) Use a graphical method to find the optimal value of p and hence find the best strategy for player B in the reduced game.

(6)

(d) (i) Find the value of the reduced game to player A.

(ii) State which option player A should never play in the reduced game.

(iii) Hence find the best strategy for player A in the reduced game.

(4)

(Total for Question 3 is 14 marks)

4. A sequence $\{u_n\}$, where $n \geqslant 0$, satisfies the recurrence relation

$$u_{n+1} = \frac{3}{2}u_n - 2n^2 - 4 \qquad u_0 = k$$

where k is an integer.

(a) Determine an expression for u_n in terms of n and k.

(6)

Given that $u_{10} > 5000$

(b) determine the minimum possible value of k.

(2)

(Total for Question 4 is 8 marks)

TOTAL FOR DECISION MATHEMATICS 2 IS 40 MARKS

Pearson Edexcel Level 3 GCE

Paper reference	8FM0/28

Further Mathematics

Advanced Subsidiary
Further Mathematics options
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Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ▶

Answer ALL questions. Write your answers in the answer book provided.

1. Four workers, A, B, C and D, are each to be assigned to one of four tasks, P, Q, R and S.

Each worker must be assigned to one task, and each task must be done by exactly one worker.

Worker C cannot be assigned to task Q and worker D cannot be assigned to task S.

The time, in minutes, that each worker takes to complete each task is shown in the table below.

	P	Q	R	S
A	54	48	51	52
B	55	51	53	58
C	52	–	53	54
D	67	63	68	–

The Hungarian algorithm is to be used to find the minimum total time for the four workers to complete the tasks.

(a) Modify the table so that the Hungarian algorithm may be used.

(1)

(b) Reducing rows first, use the Hungarian algorithm to obtain an allocation that minimises the total time. You should explain how any initial row and column reductions are made and also how you determine if the table is optimal at each stage.

(6)

(Total for Question 1 is 7 marks)

2.

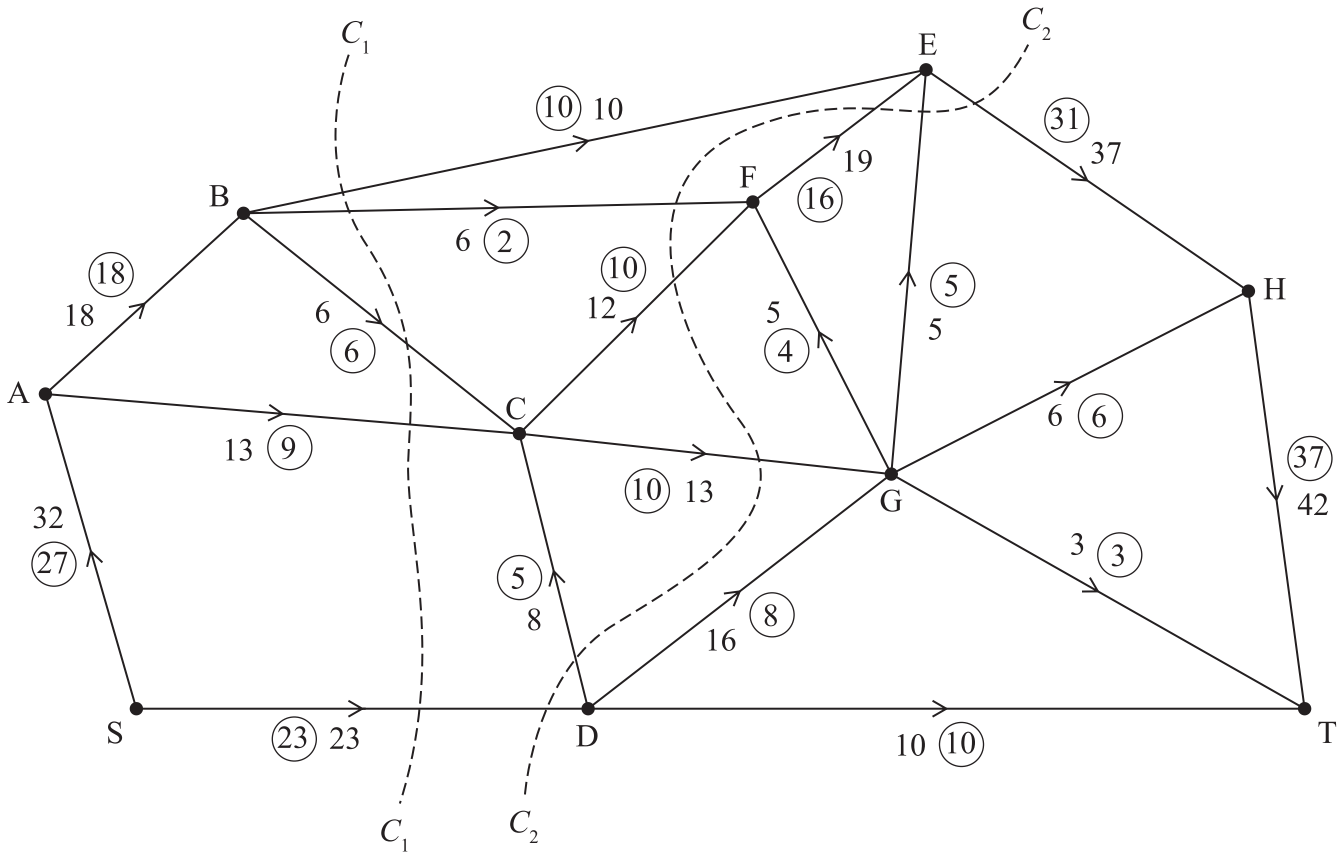

Figure 1

Figure 1 shows a capacitated, directed network of pipes. The number on each arc represents the capacity of the corresponding pipe. The numbers in circles represent a feasible flow from S to T.

(a) State the value of this flow.
(1)

(b) List the eight saturated arcs.
(1)

(c) Explain why arc EH can never be full to capacity.
(1)

(d) Find the capacity of

(i) cut C_1

(ii) cut C_2
(2)

(e) Write down a flow-augmenting route that increases the flow by three units.
(1)

Given that the flow through the network is increased by three units,

(f) prove that this new flow is maximal.
(3)

(Total for Question 2 is 9 marks)

3. Terry and June play a zero-sum game. The pay-off matrix shows the number of points that Terry scores for each combination of strategies.

		June	
		Option X	Option Y
Terry	Option A	1	4
	Option B	–2	6
	Option C	–1	5
	Option D	8	–4

(a) Explain the meaning of 'zero-sum' game.

(1)

(b) Verify that there is no stable solution to the game.

(2)

(c) Write down the pay-off matrix for June.

(1)

(d) (i) Find the best strategy for June, defining any variables you use.

(ii) State the value of the game to Terry.

(7)

Let Terry play option A with probability t.

(e) By writing down a linear equation in t, find the best strategy for Terry.

(3)

(Total for Question 3 is 14 marks)

4. A sequence $\{u_n\}$, where $n \geqslant 0$, satisfies the recurrence relation

$$u_{n+1} + 3u_n = n + k$$

where k is a non-zero constant.

Given that $u_0 = 1$

(a) solve the recurrence relation, giving u_n in terms of k and n.

(7)

Given that u_n is a linear function of n,

(b) use your answer to part (a) to find the value of u_{100}

(3)

(Total for Question 4 is 10 marks)

TOTAL FOR DECISION MATHEMATICS 2 IS 40 MARKS

END

Pearson Edexcel Level 3 GCE

Paper reference	8FM0/28

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- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.
- Good luck with your examination.

Turn over ▶

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P 6 6 7 9 5 A

Pearson

Write your answers in the answer book provided.

1. Five workers, A, B, C, D and E, are available to complete four tasks, P, Q, R and S.

Each task must be assigned to exactly one worker and each worker can do at most one task.

Worker B cannot be assigned to task R.

The amount, in pounds, that each worker will earn if they are assigned to each task is shown in the table below.

	P	**Q**	**R**	**S**
A	55	56	58	57
B	60	61	–	64
C	59	60	62	63
D	64	66	71	69
E	65	68	72	66

The Hungarian algorithm is to be used to find the maximum total amount that can be earned by the five workers.

(a) Explain how the table should be modified to allow the Hungarian algorithm to be used, giving reasons for your answer.

(2)

(b) Reducing rows first, use the Hungarian algorithm to obtain the maximum possible total earnings. You should explain how any initial row and column reductions were made and how you determined if the table was optimal at each stage.

(7)

(Total for Question 1 is 9 marks)

2.

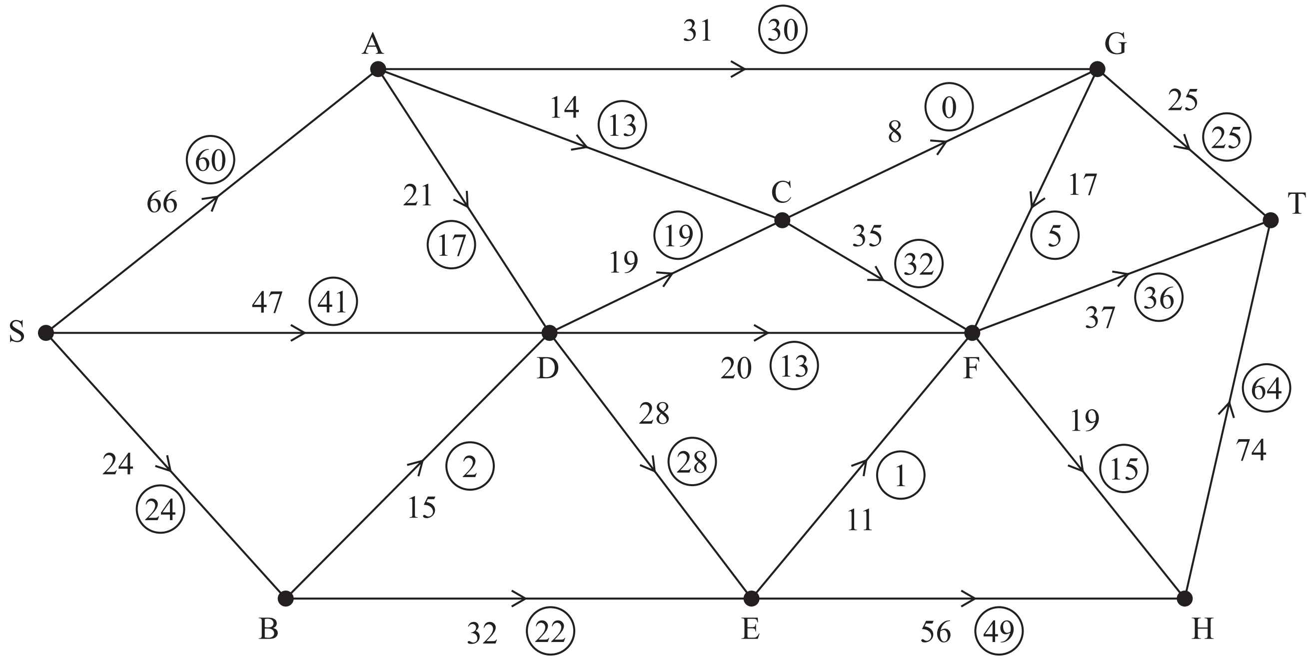

Figure 1

Figure 1 shows a capacitated, directed network. The number on each arc represents the capacity of that arc. The numbers in circles represent an initial flow.

(a) State the value of the initial flow.

(1)

(b) Obtain the capacity of the cut that passes through the arcs AG, CG, GF, FT, FH and EH.

(1)

(c) Complete the initialisation of the labelling procedure on Diagram 1 in the answer book by entering values along SD, BD, BE and GF.

(2)

(d) Use the labelling procedure to find a maximum flow through the network. You must list each flow-augmenting route you use, together with its flow.

(3)

(e) Use the answer to part (d) to add a maximum flow pattern to Diagram 2 in the answer book.

(1)

(f) Prove that your answer to part (e) is optimal.

(3)

(Total for Question 2 is 11 marks)

3. **In your answer to this question you must show detailed reasoning.**

A two-person zero-sum game is represented by the following pay-off matrix for player A.

	B plays X	B plays Y
A plays Q	4	−3
A plays R	2	−1
A plays S	−3	5
A plays T	−1	3

(a) Verify that there is no stable solution to this game.

(2)

Player B plays their option X with probability p.

(b) Use a graphical method to find the optimal value of p and hence find the best strategy for player B.

(6)

(c) Find the value of the game to player A.

(1)

(d) Hence find the best strategy for player A.

(2)

(Total for Question 3 is 11 marks)

4. Sarah takes out a mortgage of £155 000 to buy a house. Interest is added each month on the outstanding balance at a constant rate of r % each month. Sarah makes fixed monthly repayments to reduce the amount owed.

Each month, interest is added, and then her monthly repayment is used to reduce the outstanding amount owed.

The recurrence relationship for the amount of the mortgage outstanding after $n + 1$ months is modelled by

$$u_{n+1} = 1.0025u_n - x \qquad n \geqslant 0$$

where £u_n is the amount of the mortgage outstanding after n months and £x is the monthly repayment.

(a) State the value of r.

(1)

(b) Solve the recurrence relation to find an expression for u_n in terms of x and n.

(5)

Given that the mortgage will be paid off in exactly 30 years,

(c) determine, to 2 decimal places, the least possible value of x.

(3)

(Total for Question 4 is 9 marks)

TOTAL FOR DECISION MATHEMATICS 2 IS 40 MARKS

END

Pearson Edexcel Level 3 GCE

Thursday 08 October 2020

Afternoon | Paper Reference **8FM0/28**

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Further Mathematics options
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Turn over ▶

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P 6 2 6 7 5 A 0 1 0 4

Pearson

1.

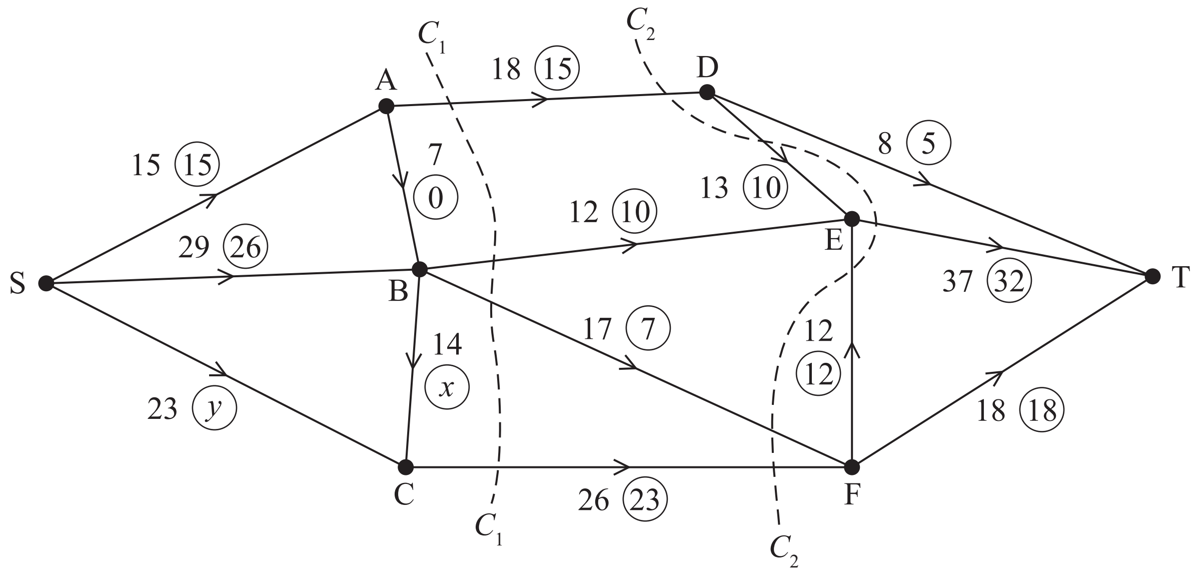

Figure 1

Figure 1 shows a capacitated, directed network of pipes. The number on each arc represents the capacity of the corresponding pipe. The numbers in circles represent a feasible flow from S to T.

(a) (i) Find the value of x.

(ii) Find the value of y.

(2)

(b) List the saturated arcs.

(1)

Two cuts, C_1 and C_2, are shown in Figure 1.

(c) Find the capacity of

(i) C_1

(ii) C_2

(2)

(d) Write down a flow-augmenting route, using the arc CF, that increases the flow by two units.

(1)

Given that the flow through the network is increased by two units using the route found in (d),

(e) prove that this new flow is maximal.

(3)

(Total for Question 1 is 9 marks)

2. Four workers, A, B, C and D, are each to be assigned to one of four tasks, P, Q, R and S.

Each worker must be assigned to one task, and each task must be done by exactly one worker.

Worker C cannot be assigned to task Q.

The amount, in pounds, that each worker would earn when assigned to each task is shown in the table below.

	P	Q	R	S
A	72	98	59	84
B	67	87	68	86
C	70	–	62	79
D	78	93	64	81

The Hungarian algorithm is to be used to find the maximum total amount that can be earned by the four workers.

(a) Explain how the table should be modified so that the Hungarian algorithm may be applied.

(2)

(b) Modify the table so that the Hungarian algorithm may be applied.

(1)

(c) Reducing rows first, use the Hungarian algorithm to obtain an allocation that maximises the total earnings. You should explain how any initial row and column reductions were made and also how you determined if the table was optimal at each stage.

(6)

(Total for Question 2 is 9 marks)

3. Two teams, A and B, each have three team members. One member of Team A will compete against one member of Team B for 10 rounds of a competition. None of the rounds can end in a draw.

Table 1 shows, for each pairing, the expected number of rounds that the member of Team A will win minus the expected number of rounds that the member of Team B will win. These numbers are the scores awarded to Team A. This competition between Teams A and B is a zero-sum game. Each team must choose one member to play. Each team wants to choose the member who will maximise its score.

		Team B		
		Paul	Qaasim	Rashid
Team A	Mischa	4	–6	2
	Noel	0	–2	6
	Olive	–6	2	0

Table 1

(a) (i) Find the number of rounds that Team A expects to win if Team A chooses Mischa and Team B chooses Paul.

(ii) Find the number of rounds that Team B expects to win if Team A chooses Noel and Team B chooses Qaasim.

(2)

Table 1 models this zero-sum game.

(b) (i) Find the play-safe strategies for the game.

(ii) Explain how you know that the game is not stable.

(4)

(c) Determine which team member Team B should choose if Team B thinks that Team A will play safe. Give a reason for your answer.

(1)

At the last minute, Rashid is ill and is therefore unavailable for selection by Team B.

(d) Find the best strategy for Team B, defining any variables you use.

(7)

(Total for Question 3 is 14 marks)

---

4. A sequence $\{u_n\}$, where $n \geqslant 1$, satisfies the recurrence relation

$$2u_n = u_{n-1} - kn^2 \quad \text{where} \quad 4u_2 - u_0 = 27k^2$$

and k is a non-zero constant.

Show that, as n becomes large, u_n can be approximated by a quadratic function of the form $an^2 + bn + c$ where a, b and c are constants to be determined.

(8)

(Total for Question 4 is 8 marks)

TOTAL FOR DECISION MATHEMATICS 2 IS 40 MARKS

END

Pearson Edexcel Level 3 GCE

Thursday 16 May 2019

Afternoon | Paper Reference **8FM0-28**

Further Mathematics

Advanced Subsidiary
Further Mathematics options
28: Decision Mathematics 2
(Part of option K only)

You must have:
Mathematical Formulae and Statistical Tables (Green), calculator,
D2 Answer Book (enclosed)

Candidates may use any calculator allowed by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of the answer book with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the Answer Book provided – *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- The total mark for this part of the examination is 40. There are 4 questions.
- The marks for each question are shown in brackets – *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over

P61640A

1/1/1/1/1/1/

1. Three workers, A, B and C, are each to be assigned to one of four tasks, P, Q, R and S.

Each worker must be assigned to at most one task, and each task must be done by at most one worker.

The amount, in pounds, that each worker will earn while assigned to each task is shown in the table below.

	P	Q	R	S
A	32	40	37	42
B	29	32	35	41
C	37	33	39	40

The Hungarian algorithm is to be used to find the maximum total amount that can be earned by the three workers.

(a) Explain how the table should be modified.

(2)

(b) (i) Reducing rows first, use the Hungarian algorithm to obtain an allocation which maximises the total earnings.

(ii) Explain how any initial row and column reductions were made and also how you determined if the table was optimal at each stage.

(7)

(Total for Question 1 is 9 marks)

2. (a) Find the general solution of the recurrence relation

$$u_{n+1} = 3u_n + 2^n \qquad n \geqslant 1$$

(4)

(b) Find the particular solution of this recurrence relation for which $u_1 = u_2$

(2)

(Total for Question 2 is 6 marks)

3.

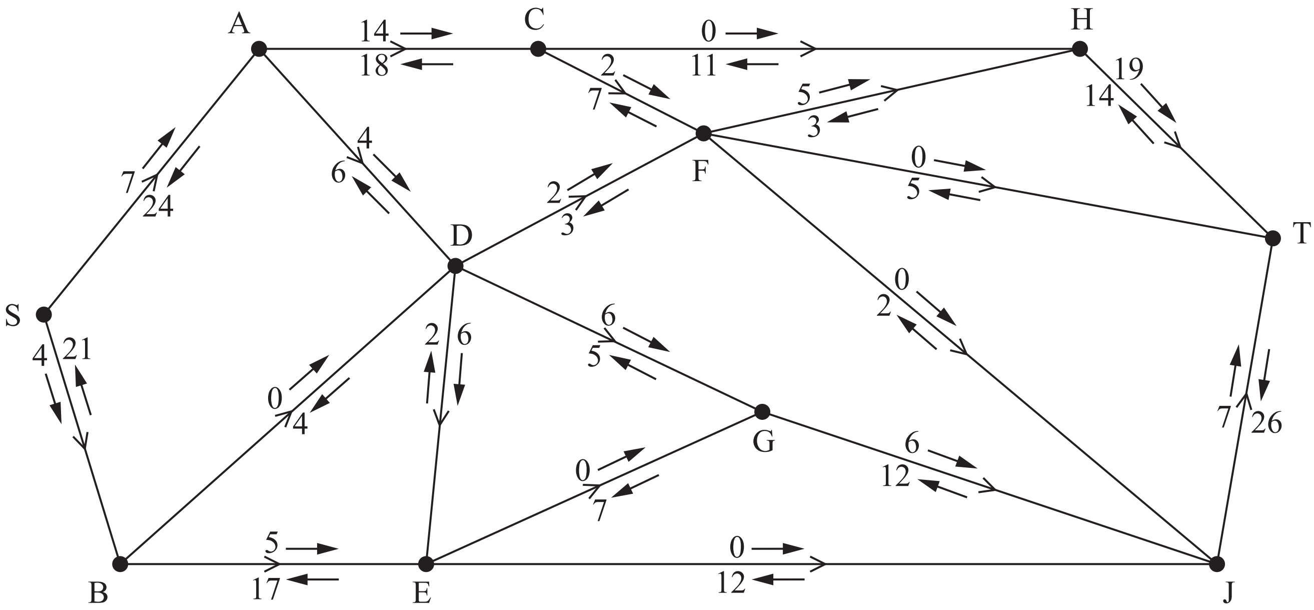

Figure 1

Alexa is monitoring a system of pipes through which fluid can flow from the source, S, to the sink, T. Currently, fluid is flowing through the system from S to T.

Alexa initialises the labelling procedure for this system, and the excess capacities and potential backflows are shown on the arrows either side of each arc, as shown in Figure 1.

(a) State the value of the initial flow.

(1)

(b) Explain why arcs DF and DG can never both be full to capacity.

(1)

(c) Obtain the capacity of the cut that passes through the arcs AC, AD, BD, DE, EG and EJ.

(1)

(d) Use the labelling procedure to find a maximum flow through the network. You must list each flow-augmenting route you use, together with its flow.

(3)

(e) Use your answers to part (d) to find a maximum flow pattern for this system of pipes and draw it on Diagram 1 in the answer book.

(1)

(f) Prove that the answer to part (e) is optimal.

(3)

(Total for Question 3 is 10 marks)

4. The table below gives the pay-off matrix for a zero-sum game between two players, Aljaz and Brendan. The values in the table show the pay-offs for Aljaz.

		Brendan		
		Option X	Option Y	Option Z
Aljaz	Option P	−6	−1	2
	Option Q	5	4	−7
	Option R	5	6	3

(a) (i) Show that this game is stable.

(ii) State the value of this game to Brendan.

(3)

Option R is removed from Aljaz's choices and the reduced game, with option R removed, is no longer stable.

(b) Find the best strategy for Aljaz in this reduced game, defining any variable you use.

(7)

(c) Explain why Brendan should never play option Y

(1)

Let Brendan play option X with probability q

(d) (i) Explain why q satisfies the equation $6q - 2(1 - q) = 1.6$

(ii) Hence find the best strategy for Brendan in this reduced game.

(4)

(Total for Question 4 is 15 marks)

TOTAL FOR DECISION MATHEMATICS 2 IS 40 MARKS

END

Pearson Edexcel Level 3 GCE

Further Mathematics

Advanced Subsidiary
Further Mathematics options
28: Decision Mathematics 2
(Part of option K only)

Thursday 17 May 2018 – Afternoon

Paper Reference
8FM0-28

You must have:
Mathematical Formulae and Statistical Tables, calculator,
D2 Answer Book (enclosed)

Candidates may use any calculator allowed by the regulations of the Joint Council for Qualifications. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of the answer book with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the Answer Book provided – *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- The total mark for this part of the examination is 40. There are 4 questions.
- The marks for each question are shown in brackets – *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over

P60209A

1/1/1/

P 6 0 2 0 9 A

Pearson

Answer ALL questions. Write your answers in the Answer Book provided.

1. Four workers, A, B, C and D, are to be assigned to four tasks, P, Q, R and S. Each worker must be assigned to exactly one task and each task must be done by only one worker. The time, in hours, that each worker takes to complete each task is shown in the table below.

	P	Q	R	S
A	7.5	3.5	8	9.5
B	5	2	7	7.5
C	4	3.5	3.5	8
D	6	5	3.5	4

Reducing rows first, use the Hungarian algorithm to obtain an allocation which minimises the total time. You must explain your method and show the table after each stage.

(Total for Question 1 is 5 marks)

2. (a) Explain what the term 'zero-sum game' means.

(1)

Two teams, A and B, are to face each other as part of a quiz.

There will be several rounds to the quiz with 10 points available in each round.

For each round, the two teams will each choose a team member and these two people will compete against each other until all 10 points have been awarded. The number of points that Team A can expect to gain in each round is shown in the table below.

		Team B		
		Paul	Qaasim	Rashid
Team A	Mischa	5	6	3
	Noel	4	1	7
	Olive	4	5	8

The teams are each trying to maximise their number of points.

(b) State the number of points that Team B will expect to gain each round if Team A chooses Noel and Team B chooses Rashid.

(1)

(c) Explain why subtracting 5 from each value in the table will model this situation as a zero-sum game.

(1)

(d) (i) Find the play-safe strategies for the zero-sum game.

(ii) Explain how you know that the game is not stable.

(4)

At the last minute, Olive becomes unavailable for selection by Team A.
Team A decides to choose its player for each round so that the probability of choosing Mischa is p and the probability of choosing Noel is $1 - p$.

(e) Use a graphical method to find the optimal value of p for Team A and hence find the best strategy for Team A.

(6)

For this value of p,

(f) (i) find the expected number of points awarded, per round, to Team A,

(ii) find the expected number of points awarded, per round, to Team B.

(2)

(Total for Question 2 is 15 marks)

3.

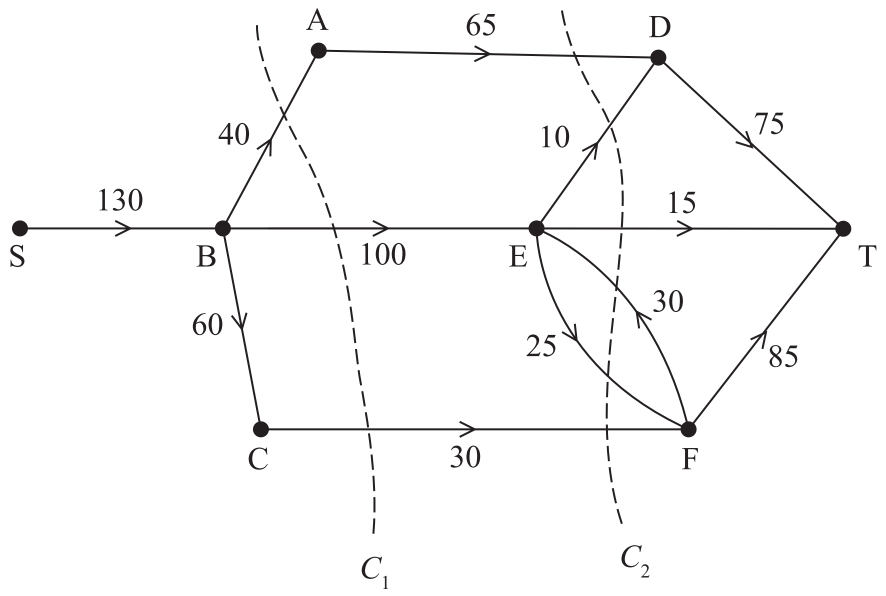

Figure 1

Figure 1 models the flow of fluid through a system of pipes from a source, S, to a sink, T. The weights on the arcs show the capacities of the corresponding pipes in litres per minute. Two cuts C_1 and C_2 are shown.

(a) Find the capacity of

(i) cut C_1

(ii) cut C_2

(2)

(b) Using only the capacities of cuts C_1 and C_2 state what can be deduced about the maximum possible flow through the system.

(1)

(c) On Diagram 1 in the answer book, show how a flow of 120 litres per minute from S to T can be achieved. You do not need to apply the labelling procedure to find this flow.

(2)

(d) Prove that 120 litres per minute is the maximum possible flow through the system.

(2)

A new pipe is planned from S to A. Let the capacity of this pipe be x litres per minute.

(e) Find, in terms of x where necessary, the maximum possible flow through the new system.

(3)

(Total for Question 3 is 10 marks)

4. A village has an expected population growth rate (birth rate minus death rate) of r% per year.
In addition, N people are expected to move into the village each year.
The expected population of the village is modelled by

$$u_{n+1} = 1.02\,u_n + 50,$$

where u_n is the expected population of the village n years from now.

(a) State

(i) the value of r,

(ii) the value of N.

(2)

Given that the population 1 year from now is expected to be 560

(b) solve the recurrence relation for u_n

(5)

(c) Hence determine, using algebra, the number of years from now when the model predicts that the population of the village will first be greater than 3000

(3)

(Total for Question 4 is 10 marks)

TOTAL FOR DECISION MATHEMATICS 2 IS 40 MARKS

END

Pearson Edexcel Level 3 GCE

Further Mathematics

Advanced Subsidiary
Further Mathematics options
Paper 2K: Decision Mathematics 1 and Decision Mathematics 2

Sample Assessment Material for first teaching September 2017 **Time: 1 hour 40 minutes**	Paper Reference **8FM0/2K**

You must have:
Decision Mathematics Answer Book (enclosed), calculator

Candidates may use any calculator permitted by Pearson regulations. Calculators must not have the facility for algebraic manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If a pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- Write your answers for this paper in the Decision Mathematics answer book provided.
- There are **two** sections in this question paper. Answer **all** the questions in Section A and **all** the questions in Section B.
- Answer the questions in the spaces provided – *there may be more space than you need*.
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Answers should be given to three significant figures unless otherwise stated.
- Do not return this question paper with the answer book.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 9 questions in this question paper. The total mark for this paper is 80.
- The marks for **each** question are shown in brackets – *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over

S58536A

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S 5 8 5 3 6 A 0 1 1 1

Pearson

SECTION B

Answer ALL questions. Write your answers in the answer book provided.

6. Six workers, A, B, C, D, E and F, are to be assigned to five tasks, P, Q, R, S and T.

Each worker can be assigned to at most one task and each task must be done by just one worker.

The time, in minutes, that each worker takes to complete each task is shown in the table below.

	P	Q	R	S	T
A	32	32	35	34	33
B	28	35	31	37	40
C	35	29	33	36	35
D	36	30	34	33	35
E	30	31	29	37	36
F	29	28	32	31	34

Reducing rows first, use the Hungarian algorithm to obtain an allocation which minimises the total time. You must explain your method and show the table after each stage.

(Total for Question 6 is 9 marks)

7. In two-dimensional space, lines divide a plane into a number of different regions.

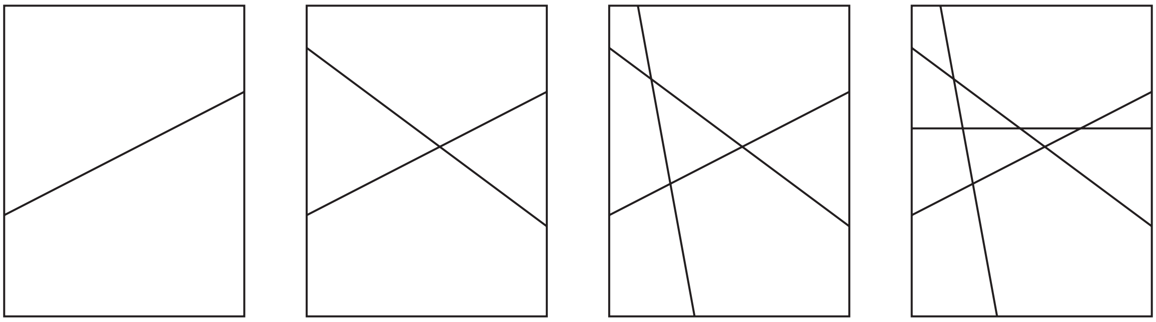

Figure 1 **Figure 2** **Figure 3** **Figure 4**

It is known that:

- One line divides a plane into 2 regions, as shown in Figure 1
- Two lines divide a plane into a maximum of 4 regions, as shown in Figure 2
- Three lines divide a plane into a maximum of 7 regions, as shown in Figure 3
- Four lines divide a plane into a maximum of 11 regions, as shown in Figure 4

(a) Complete the table in the answer book to show the maximum number of regions when five, six and seven lines divide a plane.

(1)

(b) Find, in terms of u_n, the recurrence relation for u_{n+1}, the maximum number of regions when a plane is divided by $(n + 1)$ lines where $n \geqslant 1$

(1)

(c) (i) Solve the recurrence relation for u_n

(ii) Hence determine the maximum number of regions created when 200 lines divide a plane.

(3)

(Total for Question 7 is 5 marks)

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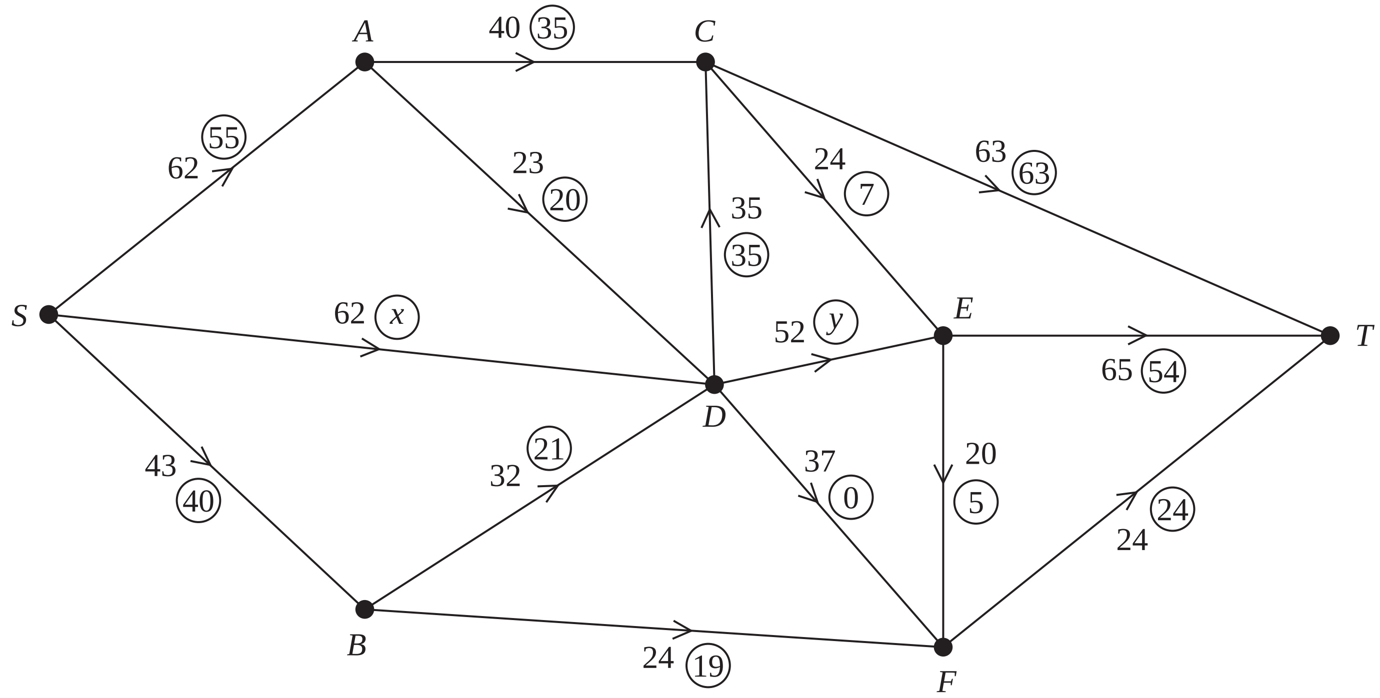

Figure 5

Figure 5 represents a network of corridors in a school. The number on each arc represents the maximum number of students, per minute, that may pass along each corridor at any one time. At 11 am on Friday morning, all students leave the hall (S) after assembly and travel to the cybercafé (T). The numbers in circles represent the initial flow of students recorded at 11 am one Friday.

(a) State an assumption that has been made about the corridors in order for this situation to be modelled by a directed network.

(1)

(b) Find the value of x and the value of y, explaining your reasoning.

(3)

Five new students also attend the assembly in the hall the following Friday. They too need to travel to the cybercafé at 11 am. They wish to travel together so that they do not get lost. You may assume that the initial flow of students through the network is the same as that shown in Figure 5 above.

(c) (i) List all the flow augmenting routes from S to T that increase the flow by at least 5

(ii) State which route the new students should take, giving a reason for your answer.

(3)

(d) Use the answer to part (c) to find a maximum flow pattern for this network and draw it on Diagram 1 in the answer book.

(1)

(e) Prove that the answer to part (d) is optimal.

(3)

The school is intending to increase the number of students it takes but has been informed it cannot do so until it improves the flow of students at peak times. The school can widen corridors to increase their capacity, but can only afford to widen one corridor in the coming term.

(f) State, explaining your reasoning,

(i) which corridor they should widen,

(ii) the resulting increase of flow through the network.

(3)

(Total for Question 8 is 14 marks)

9. A two person zero-sum game is represented by the following pay-off matrix for player *A*.

	***B* plays 1**	***B* plays 2**	***B* plays 3**
***A* plays 1**	4	1	2
***A* plays 2**	2	4	3

(a) Verify that there is no stable solution.

(3)

(b) (i) Find the best strategy for player *A*.

(ii) Find the value of the game to her.

(9)

(Total for Question 9 is 12 marks)

TOTAL FOR SECTION B IS 40 MARKS
TOTAL FOR PAPER IS 80 MARKS

Pearson Edexcel Level 3 GCE

Specimen Paper

Paper Reference **8FM0/28**

Further Mathematics
Advanced Subsidiary
28: Decision Mathematics 2

You must have:
Mathematical Formulae and Statistical Tables, calculator,
D2 Answer Book (enclosed)

Total Marks

Candidates may use any calculator allowed by the regulations of the Joint Council for Qualifications. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of the answer book with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the Answer Book provided – *there may be more space than you need*.
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- The total mark for this part of the examination is 40. There are 4 questions.
- The marks for each question are shown in brackets – *use this as a guide as to how much time to spend on each question*.

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ▶

Pearson

Answer ALL questions. Write your answers in the answer book provided.

1. The table below is a cost matrix for an allocation problem. The total cost is to be minimised.

Each of workers A, B, C and D must be assigned to exactly one of tasks P, Q, R and S.

It is not possible to assign B to R or C to S.

	P	Q	R	S
A	7	8	18	6
B	8	21	–	12
C	6	17	9	–
D	16	20	13	11

(a) The table must be adapted before the Hungarian algorithm can be applied.

(i) Explain how to adapt the table.

(ii) Explain why this adaptation allows the algorithm to be used to solve the problem.

(2)

(b) Reducing rows first, use the Hungarian algorithm to obtain an allocation which minimises the total cost. You must

(i) show the table after each stage,

(ii) explain how you know whether you have reached an optimal result,

(iii) state the final minimum cost.

(6)

(Total for Question 1 is 8 marks)

2. (a) In the context of game theory, explain what the term "play safe strategy" means.

(2)

A two person zero-sum game is represented by the following pay-off matrix for player A.

	B plays 1	B plays 2
A plays 1	−4	3
A plays 2	4	−2
A plays 3	2	−1

(b) Verify that there is no stable solution for this game.

(2)

(c) (i) Find the best strategy for player B, defining any variables you use.

(ii) Find the value of the game to player A.

(8)

(Total for Question 2 is 12 marks)

3.

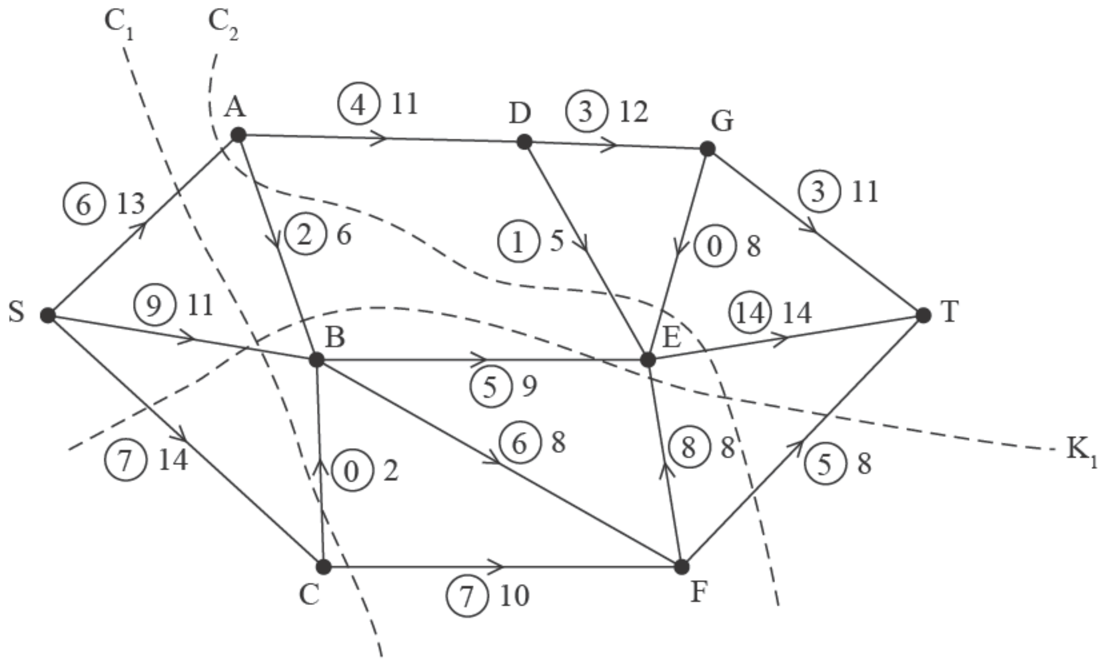

Figure 1

Figure 1 shows a capacitated, directed network. The number on each arc is the capacity of that arc. The numbers in circles represent an initial flow from S to T.

(a) Explain why the dotted line marked K_1 in Figure 1 is not a cut.

(1)

Two cuts C_1 and C_2 are shown in Figure 1.

(b) Write down

(i) the capacity of each of the two cuts C_1 and C_2

(ii) the value of the initial flow.

(3)

(c) Complete the initialisation of the labelling procedure on Diagram 1 in the answer book by entering values along arcs SA, AD, DE, GE and GT.

(2)

(d) Hence use the labelling procedure to find a maximal flow pattern for the network.
You must

(i) list each flow-augmenting path you use, together with its flow,

(ii) show your maximal flow pattern on Diagram 2 in the answer book.

(4)

(e) (i) State the value of the maximal flow,

(ii) prove that your flow is maximal.

(3)

(Total for Question 3 is 13 marks)

4. Rebecca invests a sum of money in a savings account which earns 3% interest per annum. At the end of each year, she withdraws £600 to pay for her grandson's music lessons for the following year.

Let x_n represent the amount remaining in the account at the end of the nth year, after the interest for the year has been paid and £600 has been withdrawn.

This information is represented by the recurrence relation

$$x_n = 1.03x_{n-1} - 600 \qquad n \geqslant 1$$

(a) Find a general solution for this recurrence relation.

(3)

Given that Rebecca invested £4000 into the account at the start of the first year,

(b) use this information to find a particular solution for this recurrence relation.

(1)

(c) Find the year in which the funds in the account at the end of the year fall below £600

(3)

(Total for Question 4 is 7 marks)

TOTAL FOR DECISION MATHEMATICS 2 IS 40 MARKS

END